

Causal Inference for Complex Longitudinal Data

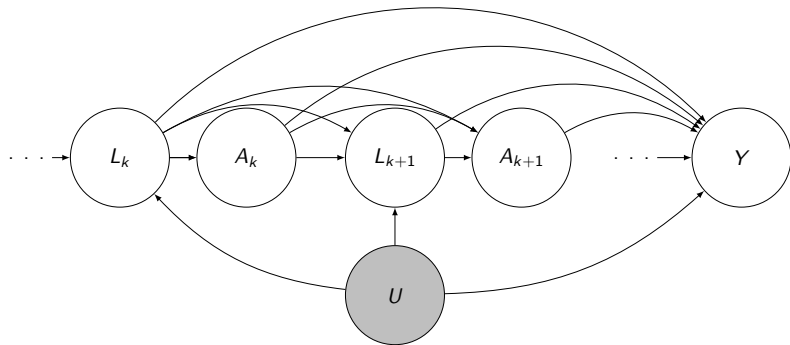
- Marginal Structural Models
- Structural Nested Models
- MSM vs. SNM

- Consider the effect of a time-varying dichotomous treatment on a continuous outcome Y measured at the end of study at time $K + 1$.
- Treatment history $\bar{A} = \bar{A}_K$, where $\bar{A}_k = (A_0, A_1, \dots, A_k)$.
- Covariates history $\bar{L} = \bar{L}_K$, where $\bar{L}_k = (L_0, L_1, \dots, L_k)$.

Identifying assumptions

- **Consistency:** If $\bar{A} = \bar{a}$, then $Y = Y^{\bar{a}}$.
- **Sequential randomization:** $Y^{\bar{a}} \perp A_k | \bar{A}_{k-1} = \bar{a}_{k-1}, \bar{L}_k$ for all possible values $\bar{a}$.
- **Positivity:** If $f_{\bar{A}_{k-1}, \bar{L}_k}(\bar{a}_{k-1}, \bar{l}_k) \neq 0$, then we have $f_{A_k | \bar{A}_{k-1}, \bar{L}_k}(a_k | \bar{a}_{k-1}, \bar{l}_k) \neq 0$ for all a_k .

Causal DAGs



Under the three assumptions, with covariate L_t discrete, the g-formula for $E(Y^{\bar{a}})$ is

$$E(Y^{\bar{a}}) = \sum_{\bar{l}} E(Y | \bar{A} = \bar{a}, \bar{L} = \bar{l}) \prod_{k=0}^K f(l_k | \bar{A}_{k-1} = \bar{a}_{k-1}, \bar{L}_{k-1} = \bar{l}_{k-1})$$

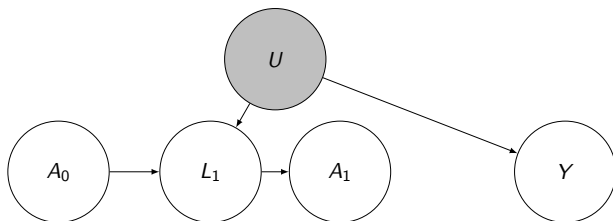
When covariate L_t is continuous, the sum is replaced by integral in the g-formula.

Take 2 stages as an example.

$$\begin{aligned} & E(Y^{a_0, a_1}) \\ &= \int E(Y^{a_0, a_1} | L_0 = l_0) f_{L_0}(l_0) dl_0 \\ &= \int E(Y^{a_0, a_1} | L_0 = l_0, A_0 = a_0) f_{L_0}(l_0) dl_0 \\ &= \int \int E(Y^{a_0, a_1} | L_0 = l_0, A_0 = a_0, L_1 = l_1) \times \\ &\quad f_{L_1|A_0, L_0}(l_1 | A_0 = a_0, L_0 = l_0) f_{L_0}(l_0) dl_1 dl_0 \\ &= \int \int E(Y | L_0 = l_0, A_0 = a_0, L_1 = l_1, A_1 = a_1) \times \\ &\quad f_{L_1|A_0, L_0}(l_1 | A_0 = a_0, L_0 = l_0) f_{L_0}(l_0) dl_1 dl_0 \end{aligned}$$

Null paradox

The main issue of specifying models of g-formula is that under standard parametrization, there is no parameter to encode the null hypothesis of no joint effect of (a_0, a_1) .



Null paradox

- Suppose that L_1 is binary and Y is continuous, so that the g-formula in this graph gives

$$E(Y^{a_0, a_1}) = \sum_{l_1=0}^1 E(Y|A_0 = a_0, A_1 = a_1, L_1 = l_1) \times f_{L_1|A_0}(l_1|A_0 = a_0)$$

- A standard modeling approach would fit a linear regression

$$E(Y|A_0 = a_0, A_1 = a_1, L_1 = l_1; \gamma) = (1, a_0, a_1, l_1)\gamma$$

and a logistic regression

$$\text{logit}\{f(L_1 = 1|A_0 = a_0; \alpha)\} = (1, a_0)\alpha$$

where $\gamma = (\gamma_0, \gamma_1, \gamma_2, \gamma_3)^\top$ and $\alpha = (\alpha_0, \alpha_1)^\top$.

- The counterfactual mean is

$$\begin{aligned} E(Y^{a_0, a_1}) \\ &= \sum_{l_1=0}^1 E(Y|A_0 = a_0, A_1 = a_1, L_1 = l_1; \gamma) \times f(l_1|A_0 = a_0; \alpha) \\ &= \left(1, a_0, a_1, \frac{\exp((1, a_0)\alpha)}{1 + \exp((1, a_0)\alpha)}\right) \gamma \end{aligned}$$

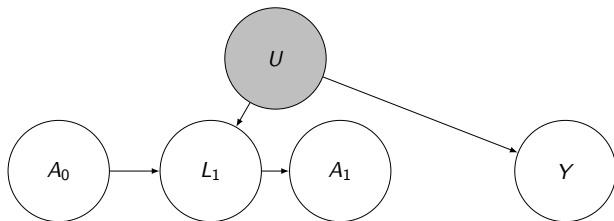
- The counterfactual mean is

$$\begin{aligned} E(Y^{a_0, a_1}) &= \sum_{l_1=0}^1 E(Y|A_0 = a_0, A_1 = a_1, L_1 = l_1; \gamma) \times f(l_1|A_0 = a_0; \alpha) \\ &= \left(1, a_0, a_1, \frac{\exp((1, a_0)\alpha)}{1 + \exp((1, a_0)\alpha)}\right) \gamma \end{aligned}$$

- $E(Y^{a_0, a_1})$ does not depend on (a_0, a_1) if either $\gamma_1 = \gamma_2 = \gamma_3 = 0$ or $\gamma_1 = \gamma_2 = \alpha_1 = 0$

Null paradox

- Recall that



$$E(Y|A_0 = a_0, A_1 = a_1, L_1 = l_1; \gamma) = (1, a_0, a_1, l_1)\gamma$$
$$\text{logit}\{f(L_1 = 1|A_0 = a_0; \alpha)\} = (1, a_0)\alpha$$

- We have

$$\gamma_3 \neq 0$$

$$\alpha_1 \neq 0$$

Marginal structural models

- An alternative is to directly specify a model for the marginal mean $E(Y^{a_0, a_1}; \psi)$ with finite dimensional parameter ψ .

For example, we could specify

$$E(Y^{a_0, a_1}; \psi) = \psi_0 + \psi_1 a_0 + \psi_2 a_1 \text{ or}$$

$$E(Y^{a_0, a_1}; \psi) = \psi_0 + \psi_1 (a_0 + a_1) \text{ or}$$

$$E(Y^{a_0, a_1}; \psi) = \psi_0 + \psi_1 a_1$$

- Note here that ψ has a causal interpretation, it is the parameter of a Marginal Structural Mean Model (MSMM)

Marginal structural models

- MSMs focus on $E\left(Y^{\bar{A}=\bar{a}}\right)$ for all possible values $\bar{a}$.
- It is said that the time-varying treatment has a causal effect on the average value of Y if $E\left(Y^{\bar{a}}\right) - E\left(Y^{\bar{a}'}\right) \neq 0$ for at least two values $\bar{a}, \bar{a}'$.
- For the first example on the previous page,
 $E\left(Y^{a_0, a_1}; \psi\right) = E\left(Y^{0,0}; \psi\right) \iff \psi_1 = \psi_2 = 0$.

Under the three assumptions, for the following mean MSM,

$$E(Y^{\bar{a}}) = \mu_{\psi}(\bar{a}),$$

we have

$$E \left[h(\bar{A})(Y - \mu_{\psi}(\bar{A})) / W \right] = 0,$$

where

$$W = \prod_{k=0}^K W_k = \prod_{k=0}^K \frac{f(A_k | \bar{L}_k, \bar{A}_{k-1})}{f^{\star}(A_k | \bar{A}_{k-1})}.$$

Under the three assumptions, the IPTW formula for $E(Y^{\bar{a}})$ is the mean of Y among the subset $(\bar{A} = \bar{a})$ in a pseudo-population, constructed by weighting each subject by their subject-specific weights

$$W = \prod_{k=0}^K f(A_k | \bar{A}_{k-1}, \bar{L}_k)$$

or stabilized weights

$$SW = \prod_{k=0}^K \frac{f(A_k | \bar{A}_{k-1}, \bar{L}_k)}{f^*(A_k | \bar{A}_{k-1})}$$

IPTW creates a pseudo-population, in which

- the mean of $Y^{\bar{a}}$ is identical to that in the true population
- the treatment at each time t depends at most on past treatment history.

The difference is that in the unstabilized pseudo-population $P_{ps}(A_k = 1 | \bar{A}_{k-1}, \bar{L}_k) = \frac{1}{2}$, while in the stabilized pseudo-population $P_{ps}(A_k = 1 | \bar{A}_{k-1}, \bar{L}_k)$ is equal to $P(A_k = 1 | \bar{A}_{k-1})$ in the true population, where the subscript ps refers to the pseudo-population. Hence, $E(Y^{\bar{a}})$ in the true population is $E_{ps}(Y | \bar{A} = \bar{a})$.

Procedure

- Specify models for $f(A_0|L_0)$ and $f(A_1|L_1, A_0, L_0)$; say logistic regressions and obtain the MLEs $\hat{\alpha}_0$ and $\hat{\alpha}_1$
- For each person in the study, compute the weight $\hat{W} = \hat{f}(A_0|L_0; \hat{\alpha}_0) \hat{f}(A_1|L_1, A_0, L_0; \hat{\alpha}_1)$ which corresponds to the estimated probability of receiving the treatment you did indeed receive
- Regress Y on A_0 and A_1 using weighted least-squares with weights $\hat{W}^{-1}$

Proof sketch

$$\begin{aligned} & E(SW^{-1}h(A_0, A_1)Y) \\ &= E(SW^{-1}h(A_0, A_1)E(Y \mid A_0, A_1, L_0, L_1)) \\ &= \sum_{l_0, l_1, a_0, a_1} SW^{-1}h(a_0, a_1)E(Y \mid a_0, a_1, l_0, l_1) \\ &\quad \times f(a_1 \mid l_1, a_0, l_0)f(l_1 \mid l_0, a_0)f(a_0 \mid l_0)f(l_0) \\ &= \sum_{l_0, l_1, a_0, a_1} \frac{f^*(a_0)f^*(a_1 \mid a_0)}{f(a_0 \mid l_0)f(a_1 \mid l_1, a_0, l_0)} h(a_0, a_1)E(Y \mid a_0, a_1, l_0, l_1) \\ &\quad \times f(a_1 \mid l_1, a_0, l_0)f(l_1 \mid l_0, a_0)f(a_0 \mid l_0)f(l_0) \\ &= \sum_{a_0, a_1} f^*(a_0)f^*(a_1 \mid a_0) \sum_{l_0, l_1} E(Y \mid a_0, a_1, l_0, l_1)h(a_0, a_1) \\ &\quad \times f(l_1 \mid l_0, a_0)f(l_0) \\ &= \sum_{a_0, a_1} f^*(a_0)f^*(a_1 \mid a_0)E(Y^{a_0, a_1})h(a_0, a_1) \\ &= E(SW^{-1}h(A_0, A_1)E[Y^{A_0, A_1}]) \end{aligned}$$

For time k , denote lost-to-follow-up by $C_k = 0$, follow-up by $C_k = 1$. The stabilized weight for each subject is $SW^+ \times SW$, where

$$SW^+ = \prod_{k=0}^K \frac{Pr(C_k = 1 | \bar{C}_{k-1} = \bar{1}_{k-1}, \bar{A}_{k-1} = \bar{a}_{k-1}, \bar{L}_k = \bar{l}_k)}{Pr(C_k = 1 | \bar{C}_{k-1} = \bar{1}_{k-1}, \bar{A}_{k-1} = \bar{a}_{k-1})}$$

Under the three assumptions, for the following Cox MSM,

$$\lambda_{T\bar{a}}(t) = \lambda_0(t) \exp(m(\bar{a}(t), t, \psi)),$$

we have

$$E \int dN(t) \{W(t)\}^{-1} \left\{ h(\bar{A}, t) - \frac{E[h(\bar{A}, t) \exp(m(\bar{A}(t), t, \psi)) \{W(t)\}^{-1} I(Y \geq t)]}{E[\exp(m(\bar{A}(t), t, \psi)) \{W(t)\}^{-1} I(Y \geq t)]} \right\} = 0,$$

where

$$W(t) = \prod_{k=0}^{\lfloor t \rfloor} W_k = \prod_{k=0}^{\lfloor t \rfloor} \frac{f(A_k | \bar{L}_k, \bar{A}_{k-1}, Y \geq k)}{f^*(A_k | \bar{A}_{k-1}, Y \geq k)},$$

and $N(t) = I(Y \leq t, \delta = 1)$.

Under the three assumptions, for the following Cox MSM,

$$\lambda_{T^{\bar{a}}}(t) = \lambda_0(t) \exp(\psi a(t)),$$

we have

$$E \int dN(t) \{W(t)\}^{-1} \left\{ A(t) - \frac{E[A(t) \exp(\psi A(t)) \{W(t)\}^{-1} I(Y \geq t)]}{E[\exp(\psi A(t)) \{W(t)\}^{-1} I(Y \geq t)]} \right\} = 0,$$

where

$$W(t) = \prod_{k=0}^{\lfloor t \rfloor} W_k = \prod_{k=0}^{\lfloor t \rfloor} \frac{f(A_k | \bar{L}_k, \bar{A}_{k-1}, Y \geq k)}{f^*(A_k | \bar{A}_{k-1}, Y \geq k)},$$

and $N(t) = I(Y \leq t, \delta = 1)$.

Structural Mean Model (SMM)

Point exposure case

- **SMMs:**

$$g \{E(Y^a|L=l, A=a)\} - g \{E(Y^0|L=l, A=a)\} = \gamma^*(l, a; \psi)$$

- **Identity link with scalar covariate L :**

$$E(Y^a|L=l, A=a) - E(Y^0|L=l, A=a) = (\psi_0 + \psi_1 l) a$$

- **Weak Ignorability:**

$$Y^0 \perp A | L$$

SMM: estimation of parameters

- Construct a variable $U^*(\psi)$ whose conditional mean value equals the corresponding mean outcome that would have been seen if we had intervened to remove the treatment from that subset.

$$E(U^*(\psi) | L, A) = E(Y^0 | L, A)$$

- For the SMM with identity link, $U^*(\psi)$ is constructed as

$$U^*(\psi) = Y - \gamma^*(L, A; \psi)$$

SMM: estimation of parameters

- Under the construction of variable $U^*(\psi)$ and the weak ignorability assumption,

$$\begin{aligned} E(U^*(\psi) | L, A) &= E(Y^0 | L, A) \\ &= E(Y^0 | L) \\ &= E(U^*(\psi) | L) \end{aligned}$$

- Then, for an arbitrary function $d^*(A, L)$ of A and L ,

$$\text{Cov}(U^*(\psi), d^*(A, L) | L) = 0$$

because $E(X|Y) = E(X) \Rightarrow \text{Cov}(X, Y) = 0$

SMM: estimation of parameters

For an arbitrary function $d^*(A, L)$ of A and L ,

$$\begin{aligned} & \text{Cov}(U^*(\psi), d^*(A, L) | L) \\ &= E(U^*(\psi) d^*(A, L) | L) - E(U^*(\psi) | L) E(d^*(A, L) | L) \\ &= E[E(U^*(\psi) d^*(A, L) | L, A) | L] - E(U^*(\psi) | L) E(d^*(A, L) | L) \\ &= E[d^*(A, L) E(U^*(\psi) | L, A) | L] - E(U^*(\psi) | L) E(d^*(A, L) | L) \\ &= E[d^*(A, L) E(U^*(\psi) | L) | L] - E(U^*(\psi) | L) E(d^*(A, L) | L) \\ &= E(U^*(\psi) | L) E[d^*(A, L) | L] - E(U^*(\psi) | L) E(d^*(A, L) | L) \\ &= 0 \end{aligned}$$

SMM: estimation of parameters

Estimation of ψ in the SMM can thus be based on solving the following estimating equations based on the constraint, which set the empirical conditional covariance between $U^*(\psi)$ and arbitrary functions $d^*(A, L)$, given L , to zero.

$$0 = \sum_{i=1}^n [d^*(A_i, L_i) - E(d^*(A_i, L_i) | L_i)] [U_i^*(\psi) - E(U_i^*(\psi) | L_i)]$$

The functions $d^*(A, L)$ are arbitrary, whose dimension should be chosen the same as the dimension of ψ .

SMM: estimation of parameters

For instance, in the SMM with identity link and scalar covariate L , the choice $d^*(A_i, L_i) = (1, L_i')' A_i$ results in estimation equations

$$0 = \sum_{i=1}^n \begin{pmatrix} 1 \\ L_i \end{pmatrix} \{A_i - \hat{E}(A_i|L_i)\} \{Y_i - \hat{E}(Y_i|L_i) - (\psi_0 + \psi_1 L_i) [A_i - \hat{E}(A_i|L_i)]\}$$

from which estimates for (ψ_0, ψ_1) can be solved.

Time-varying setting

- Suppose the measurements are collected at fixed time points $t_0, t_1, \dots, t_{K+1}$. Let A_k denote the treatment provided at time t_k , for $k = 0, \dots, K$, and L_k observed covariates measured at the same time points. The outcome Y_k is measured at time t_k for $k = 1, \dots, K + 1$.
- The overline denotes all the variables from the beginning to current, and the underline denotes from the current to the end. For example, y_m , $m = 1, \dots, K + 1$, $\bar{y}_m = (y_1, \dots, y_m)$ and $\underline{y}_m = (y_m, \dots, y_{K+1})$.

Time-varying setting

- Let $Y_m^{\bar{a}_{m-1}}$ denote the outcome that would be observed at time t_m if one were to receive treatment history $\bar{a}_{m-1}$ through time t_{m-1} . Thus, the variables $Y_m^{\bar{a}_{m-1}}$, for $m = 1, \dots, K + 1$, are potential outcomes, which are linked to the observed data via the consistency assumption that $Y_m = Y_m^{\bar{a}_{m-1}}$ if $\bar{A}_{m-1} = \bar{a}_{m-1}$.

- Weak Sequential Ignorability:**

$$A_m \perp Y_{m+1}^{\bar{a}_{m-1}, 0_m} \mid \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}$$

for $m = 0, \dots, K$.

Structural nested model (SNM)

Structural nested models (SNMs) parameterize contrasts of $Y_{m+1}^{\bar{a}_m, 0_{m+1}}$ and $Y_{m+1}^{\bar{a}_{m-1}, 0_m}$ conditionally on treatment and covariate histories through t_m as

$$g \left\{ E \left(Y_{m+1}^{\bar{a}_m, 0_{m+1}} | \bar{L}_m = \bar{l}_m, \bar{A}_m = \bar{a}_m \right) \right\} - g \left\{ E \left(Y_{m+1}^{\bar{a}_{m-1}, 0_m} | \bar{L}_m = \bar{l}_m, \bar{A}_m = \bar{a}_m \right) \right\} \\ = \gamma_m^* (\bar{l}_m, \bar{a}_m; \psi)$$

for $m = 0, \dots, K$, where $\gamma_m^* (\bar{l}_m, \bar{a}_m; \psi)$ is a $(K + 1 - m)$ -dimensional function for each m .

A final outcome example

Alternatively, one may merely focus on the effect of treatment on the end-of-study outcome $Y \equiv Y_{K+1}$, in which case one obtains a SNM of the form

$$\begin{aligned} & g \left\{ E \left(Y^{\bar{a}_m, 0_{m+1}} | \bar{L}_m = \bar{l}_m, \bar{A}_m = \bar{a}_m \right) \right\} - g \left\{ E \left(Y^{\bar{a}_{m-1}, 0_m} | \bar{L}_m = \bar{l}_m, \bar{A}_m = \bar{a}_m \right) \right\} \\ &= \gamma_m^* (\bar{l}_m, \bar{a}_m; \psi) \end{aligned}$$

for $m = 0, \dots, K$, where $\gamma_m^* (\bar{l}_m, \bar{a}_m; \psi)$ is now 1-dimensional function for each m .

A two-stage example

- For instance, a linear SNM with identity link and only 2 time points ($K = 1$) would be constructed as the following three equations:

$$E \left(Y_2^{(a_0, a_1)} - Y_2^{(a_0, 0)} \mid \bar{L}_1 = \bar{l}_1, \bar{A}_1 = \bar{a}_1 \right) = (\psi_0 + \psi_1 l_1 + \psi_2 a_0) a_1, [\gamma_{1,2}]$$

$$E \left(Y_2^{(a_0, 0)} - Y_2^{(0, 0)} \mid L_0 = l_0, A_0 = a_0 \right) = (\psi_3 + \psi_4 l_0) a_0, [\gamma_{0,2}]$$

$$E \left(Y_1^{(a_0, 0)} - Y_1^{(0, 0)} \mid L_0 = l_0, A_0 = a_0 \right) = (\psi_0 + \psi_1 l_0) a_0. [\gamma_{0,1}]$$

- Here, the first equation parameterizes the effect of A_1 on Y_2 , the second parameterizes the effect of A_0 on Y_2 , and the last parameterizes the effect of A_0 on Y_1 , where all of those are conditional on covariates and treatments prior to the considered time points.

SNM: estimation of parameters

Construct a transformation $U_m^*(\psi)$ of $\underline{Y}_{m+1}$, in the sense that

$$E[U_m^*(\psi) | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}, A_m] = E[\underline{Y}_{m+1}^{\bar{a}_{m-1}, 0_m} | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}, A_m]$$

for $m = 0, \dots, K$.

Under the identity link, for each m , $U_m^*(\psi)$ can be constructed as a $(K + 1 - m)$ -dimensional vector with components

$$Y_k - \sum_{l=m}^{k-1} \gamma_{l,k}^*(\bar{L}_l, \bar{A}_l; \psi)$$

for $k = m + 1, \dots, K + 1$.

SNM: estimation of parameters

For the example of linear SNM with 2 time points ($K = 1$) and identity link, $U_m^*(\psi)$, for $m = 1, 0$,

$$U_1^*(\psi) = Y_2 - (\psi_0 + \psi_1 L_1 + \psi_2 A_0) A_1,$$

$$U_0^*(\psi) = \begin{pmatrix} Y_1 - (\psi_0 + \psi_1 L_0) A_0 \\ Y_2 - (\psi_0 + \psi_1 L_1 + \psi_2 A_0) A_1 - (\psi_3 + \psi_4 L_0) A_0 \end{pmatrix}$$

which satisfy that

$$E[U_m^*(\psi^*) | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}, A_m] = E\left[\underline{Y}_{m+1}^{\bar{a}_{m-1}, 0_m} | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}, A_m\right]$$

for $m = 1$ and 0 .

SNM: estimation of parameters

Under the weak sequential ignorability assumption,

$$A_m \perp Y_{m+1}^{\bar{a}_{m-1}, 0_m} | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}$$

and the construction of $U_m^*(\psi)$,

$$\begin{aligned} E[U_m^*(\psi) | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}, A_m] &= E[Y_{m+1}^{\bar{a}_{m-1}, 0_m} | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}, A_m] \\ &= E[Y_{m+1}^{\bar{a}_{m-1}, 0_m} | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}] \\ &= E[U_m^*(\psi) | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}] \end{aligned}$$

for $m = 0, \dots, K$.

By

$$E[U_m^*(\psi) | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}, A_m] = E[U_m^*(\psi) | \bar{L}_m, \bar{A}_{m-1} = \bar{a}_{m-1}]$$

we have that

$$\text{Cov}(U_m^*(\psi), d_m^*(\bar{L}_m, \bar{A}_m) | \bar{L}_m, \bar{A}_{m-1}) = 0$$

for each $m = 0, \dots, K$.

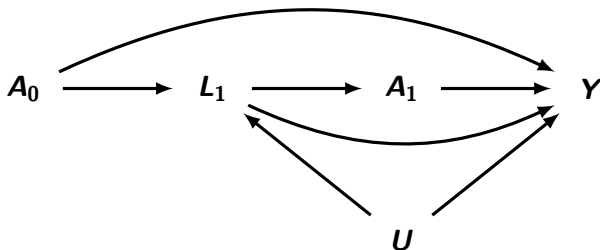
SNM: estimation of parameters

The parameter ψ can therefore be estimated by solving the corresponding empirical equation,

$$0 = \sum_{i=1}^n \sum_{m=0}^K [d_m^* (\bar{L}_{i,m}, \bar{A}_{i,m}) - E \{d_m^* (\bar{L}_{i,m}, \bar{A}_{i,m}) | \bar{L}_{i,m}, \bar{A}_{i,m-1}\}] \\ \times [U_{i,m}^* (\psi) - E \{U_{i,m}^* (\psi) | \bar{L}_{i,m}, \bar{A}_{i,m-1}\}]$$

where $d_m^* (\bar{L}_{i,m}, \bar{A}_{i,m})$, for $m = 0, \dots, K$, is an arbitrary $p \times (K + 1 - m)$ -dimensional matrix function, with p the dimension of ψ .

Parameter comparison: MSM vs. SNM



Parameter comparison: MSM vs. SNM

- The marginal structural model is given by

$$E(Y^{a_0, a_1}) = \beta_0 + \theta_0 a_0 + \theta_1 a_1 + \theta_2 a_0 a_1$$

- The structural nested model is given by

$$\begin{aligned} & E(Y^{a_0, a_1} - Y^{a_0, 0} | A_0 = a_0, L_1 = l_1, A_1 = a_1) \\ &= a_1 (\psi_1 + \psi_2 a_0 + \psi_3 l_1 + \psi_4 a_0 l_1) \\ & E(Y^{a_0, 0} - Y^{0, 0} | A_0 = a_0) = \psi_0 a_0 \end{aligned}$$

Parameter comparison: MSM vs. SNM

MSM *SNM*

$$E(Y^{1,0} - Y^{0,0}) = \theta_0 = \psi_0$$

$$E(Y^{0,1} - Y^{0,0}) = \theta_1 = \psi_1 + \psi_3 P(L_1 = 1 | A_0 = 0)$$

$$\begin{aligned} E(Y^{1,1} - Y^{1,0} - Y^{0,1} + Y^{0,0}) &= \theta_2 = \psi_2 \\ &+ \psi_3 [P(L_1 = 1 | A_0 = 1) - P(L_1 = 1 | A_0 = 0)] \\ &+ \psi_4 P(L_1 = 1 | A_0 = 1) \end{aligned}$$